USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [22CSE130]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

INTRODUCTION TO DATA SCIENCE

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 X 2=10**		
1.	**a.**	List out the areas in which Data Science can be applied.	L4	CO1	**02**
	b.	What is outliers in Dataset?	L1	CO2	**02**
	c.	What is uni-variate and bi-variate analysis?	L1	CO3	**02**
	d.	What is the main objective of machine learning?	L1	CO4	**02**
	e.	Which algorithm is best suited for recommendation systems?	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain data cleansing process with example.	L2	CO1	**05**
		OR			
	b.	Illustrate on data normalization in data processing.	L2	CO1	**05**
3.	**a.**	List the importance of data pre-processing.	L4	CO2	**05**
		OR			
	b.	Identify the different types of uni-variate tests.	L3	CO2	**05**
4.	**a.**	Summarize on the two kinds of target variables for predictive modeling.	L2	CO3	**05**
		OR			
	b.	Summarize the steps involved in feature engineering.	L2	CO2	**05**
5.	**a.**	Explain clustering and its techniques.	L2	CO4	**05**
		OR			
	b.	Explain dimension reduction in unsupervised machine learning.	L2	CO4	**05**
6.	**a.**	Analyse why model selection and evaluation are important in machine learning.	L4	CO4	**05**
		OR			
	b.	Mention the the purpose of a recommendation system.	L4	CO5	**05**
7.	**a.**	Illustrate on Social Network Analysis with example.	L4	CO5	**05**
		OR			

	b.	Explain about Text mining and sentiment analysis with an example.	L4	CO5	05
		PART-C		**6 X 10=60**	
8.	a.	Identify the components of data science. Explain the components in detail.	L3	CO1	10
		OR			
	b.	Illustrate the Real life usage of Data Science Systems.	L2	CO1	10
9.	a.	List the four types of Data analysis. Explain with an example.	L4	CO1	10
		OR			
	b.	Illustrate on Life cycle of data science process with neat sketch.	L2	CO2	10
10.	a.	Explain the concept of data pre-processing and steps involved in it with neat diagram.	L2	CO2	10
		OR			
	b.	List is the Difference between Univariate, Bivariate, and Multivariate analysis.	L4	CO3	10
11.	a.	Explain the importance of exploratory data analysis in data science.	L2	CO3	10
		OR			
	b.	Illustrate on different types of machine learning algorithm.	L2	CO4	10
12.	a.	Examine the purpose of Support Vector Machine in supervised learning technique.	L4	CO4	10
		OR			
	b.	Explain Regression Analysis and types of Regression Analysis in detail	L4	CO4	10
13.	a.	**i.** How is Data Science used in healthcare? **ii.** List the application of data science.	L1 L4	CO5	06 04
		OR			
	b.	**i.** Relate data science and social science.	L2	CO5	05
		ii. Explain about predictive analysis and time series analysis.	L2	CO5	05